MA'ADEN معادن
Saudi Arabian Mining Company شركة التعدين العربية السعودية

Project **Mansourah Massarah Gold Project**
Customer: **MA'ADEN GOLD AND BASE METALS (MA'ADEN)**

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1000 OVERALL
GEOTECHNICAL INVESTIGATION

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1.0 INTRODUCTION

This document provides the minimum requirements for geotechnical investigation as indicated in the Plant Layout with soil Test Locations and submission of a Geotechnical Report. The purpose of this guideline is to set the minimum engineering standard for geotechnical investigation for Mansourah – Massarah Gold Project. This document is intended to be used in conjunction with the applicable documents listed herein.

Larsen & Toubro Limited formed contract with Riyadh Geotechnique and Foundations (RGF), Al-Khobar for carrying out the geotechnical investigations for the proposed project site.

The investigation scheme as planned by Larsen & Toubro Limited will be carried out very shortly. Subsequently, the laboratory testing of the field retrieved samples will be conducted at RGF Geotechnical and Materials Testing Laboratories, Al Khobar.

1.1 SCOPE OF WORK

The scope of work will be as marked in the layout drawing # MD-216-1000-EG-CV-LAD-1001 & MD-216-1000-EG-CV-LAD-1002. The detailed specification for Geotechnical Investigation will be as per Ma'aden Engineering Manual for Geotechnical Investigation doc # MD-101-SMEM-EG-CV-SPC-0009. The brief summary of scope of work is listed below:

1. Surveying and Staking out of the borehole locations at the site in accordance with the locations / coordinates as indicated in layout plan drawing.

2. Drilling and sampling of twenty five (25) boreholes up to a depth of 15 m, one (01) borehole up to a depth of 8m and three (03) boreholes up to a depth of 30 m below existing ground level.

   Note: Refer Ma'aden document "MD-216-0000-CP-CT-RFP-0111, C01 BFS-Appendix 7B -Geotechnical Infrastructure" where-in borehole details for 26 nos. boreholes are already recorded and available for reference.

3. Excavation of fifty (50) test pits in soil down to 1.50m depth (4m depth for 2 TP) below existing ground level or upto rock level whichever comes first.

4. Performance of three (03) Downhole Seismic Tests down to maximum 30.0 m depth.

5. Performance of sixteen (16) Electrical Resistivity Tests at specified location using an electrode spacing of 1.0 m, 5.0 m, 10.0 m and 20.0 m (refer ASTM D G57).

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6. Performance of seven (07) plate load tests.

7. Performance of three (03) piezometer tests.

8. Performance of one (01) thermal resistivity test.

9. Backfilling of test pits and boreholes as per Ma'aden specification.

10. Performing laboratory tests to determine pertinent engineering and index properties as well as the chemical properties of the strata encountered at the project site.

11. Analyzing the field and laboratory tests data and submitting an engineering report, furnishing recommendations for the design and construction of foundations and earthwork for the proposed structures at the site.

12. The additional tests (if any) at Solar Plant area shall be decided after finalization of Solar Plant layout and its location at a later date.

1.2 INVESTIGATION METHODOLOGY

Keeping in view the scope of work and specifications for the detailed geotechnical studies, a programme of field work will be prepared. The exploratory borings will be drilled using straight rotary drilling rigs. Field tests will be performed at designated locations in accordance with relevant ASTM standards and Ma'aden specifications. Soil samples will be collected from designated depths from boreholes using appropriate samplers for identification and subsequent laboratory testing. Selected soil samples will be subjected to various laboratory tests for evaluation of classification, strength and chemical characteristics of the subsurface strata. The laboratory tests will be performed in accordance with the relevant ASTM standards and Ma'aden specification.

2.0 FIELD INVESTIGATIONS

2.1 GENERAL

The proposed field work consists of exploratory drilling and sampling of the subsurface soil samples and performance of field tests at designated locations. The borehole locations as shown in the layout plan drawing will be marked in the field by RGF.

2.2 SURVEY OF TEST LOCATIONS

The coordinates and elevation of the test locations will be measured using differential GPS by deploying RKT method.

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The coordinates for the project site are based on UTM 39/International 1924 Ellipsoid, Ain Al Abd 1970 Datum and Mean Sea Level (MSL).

2.3 EXECUTION OF BOREHOLES

The boreholes of required depth below existing ground level will be executed using straight rotary drilling method. The drilling of boreholes will be carried out using rigs equipped with automatic hammer. The subsurface stratum will be penetrated by rotary drilling method using wash boring technique. Bentonite slurry of suitable consistency will be circulated as a drilling fluid to provide stability to borehole walls. After completion of the testing, boreholes will be backfilled with a mixture of cement, bentonite, sand and water. The grout backfill materials will be proportioned to a groutable / pourable consistency and the proportions varied as per the specific requirements.

The borehole logs will be prepared in the field and a description of soil samples recovered will be shown in the respective borehole logs. Field borehole logs will be developed on the basis of the materials encountered at the site and will be later confirmed / modified by the laboratory test results.

2.4 EXCAVATION OF TEST PITS

Test pits will be excavated at the specified locations up to a maximum depth of 1.50 m below existing ground level or upto rock level whichever comes first. The test pits will be carefully logged during excavation and field logs will be developed, which will be finalized after laboratory testing.

2.5 IN-SITU SAMPLING

2.5.1 Soil Sampling

The samples recovered from split spoon sampler will be visually inspected and classified as per ASTM D-2488-09a. All the SPT will be carried out through automatic SPT hammer and samples obtained (disturbed/undisturbed) from the boreholes and test pits will be properly labelled, preserved in polythene bags and placed in plastic jars or companion jar carefully to preserve moisture content. All these samples will be carefully transported to our laboratory for further evaluation and testing.

2.5.2 Groundwater Sampling

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Observations will be made for stabilized ground water table, measured after 24 hours or more after completion of drilling of borehole. Groundwater samples will be collected from the boreholes and sent to RGF's laboratory in Al-Khobar for further testing and evaluation.

2.6 IN-SITU TESTING

2.6.1 Standard Penetration Tests

Standard Penetration Tests (SPTs) will be performed using automatic hammers at 1.0m depth intervals. These tests will be performed generally in accordance with ASTM D-1586 using a split spoon sampler of 35mm inner dia. and 50mm outer dia. The data obtained from these tests will be utilized to assess the in-situ compactness/denseness of the subsurface materials. The SPT blow counts will be recorded for a total penetration of 45 cm of SPT sampler. The number of blows required to drive the sampler through the last 30 cm viz. 'N' values will be shown on the respective borehole logs.

2.6.2 Downhole Seismic Test

Downhole seismic tests shall be conducted using a twelve channel acquisition system in the designated borehole at each 1.0m depth interval down to the depth of 30m. These tests will be performed generally in accordance with ASTM D-7400-08.

Three channels shall be used for recording P and S wave signals. One channel shall record the P wave and the other two shall record Sv and Sh. Each reading shall be measured at 1.0m vertical interval. The evaluation and interpretation of the data is done with utmost care.

Data acquisition:

For down-hole seismic survey, PVC pipe of 3.0" diameter shall be lowered in the borehole of 4.0". The annular space between the pipe and borehole wall shall be grouted with cement and bentonite grout in an appropriate ratio. After giving sufficient time for setting and curing of grout, the down-hole seismic measurements shall be taken in the PVC pipe. For this purpose a tri-axial geophone shall be lowered into the borehole and clamped at the respective depth of measurements. The tri-axial geophone shall be connected to the 12-channel seismograph.

The direct distance between the energy source at the surface and the recording geophone shall be calculated on the basis of known X, Y distances at the surface and within the bore hole. The arrival time measurements shall be recorded on the seismograph.

During the down-hole survey, accuracy of the seismograph and triggering system shall be monitored.

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2.6.3 Plate Load Test

Plate Load Tests shall be performed to evaluate the load-settlement behavior of the existing strata. These tests will be performed as per latest ASTM D-1196. Bearing plate of 60 cm diameter will be used to perform the tests. The maximum design pressure of the soil is 40 t/sqm. The maximum test load of 80 t/sqm (2 times of maximum design pressure) will be applied in equal increments of 20% of test load and each increment will be held for a minimum time of fifteen minutes. Load and deflection readings for each load increment will be recorded and this procedure will be continued until the selected total pressure will be reached.

2.6.4 Field Electrical Resistivity Tests

Electrical Resistivity Tests will be carried out at the specified locations of the site as per Standard ASTM D G 57. The measurement of the electrical resistivity of the ground will be conducted using a 4-Pin Soil Resistance Terrameter SAS 4000. The readings in the field will be taken by using Wenner's four electrode configuration.

As the depth of measurement is dependent upon electrode spacing, the readings will be taken starting from smaller electrode spacing and increased symmetrically with respect to center of electrode configuration. Electrode spacing of 1.0 m, 5.0 m, 10.0 m and 20.0 m shall be used. For each electrode spacing, apparent electrical resistivity values shall then be calculated.

3.0 LABORATORY TESTING

3.1 SOIL TESTING

Samples retrieved from boreholes will be examined in the field and then transported to RGF's laboratories for relevant laboratory testing. Laboratory testing programs will be prepared by the geotechnical engineers and will be submitted for the execution at the RGF's laboratory. The laboratory tests will be conducted on the selected representative soil samples to determine the engineering characteristics of the subsurface strata. Samples collected from the boreholes will be subjected to the following tests as per latest ASTM / BS or equivalent Standards.

- Grain size Analysis
- Atterberg Limits
- Natural Moisture Content
- Specific Gravity

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- Bulk Density
- Permeability Tests
- Relative Density
- Direct Shear Tests
- Laboratory Compaction of Soil using Modified Effort
- 3 Point Soaked CBR
- Water Absorption
- Porosity
- Unconfined Compression Test
- Point Load Strength Index
- Sulphates Content
- Chloride Content
- pH
- Total Carbonate as Calcium Carbonate

4.0 REPORTING

The deliverable will comprise the following:

Final Geotechnical Investigation Report including factual data and recommendation.

5.0 REFERENCE DOCUMENTS

a) MD-101-SMEM-EG-CV-SPC-0009 (Ma'aden Engineering Manual for Geotechnical Investigation)

C01

b) MD-216-1000-EG-CV-LAD-1001 & MD-216-1000-EG-CV-LAD-1002 (Plant Layout with Soil Test Location)

c) Ma'aden Geotechnical Investigation Report "MD-216-0000-CP-CT-RFP-0111, C01 BFS-Appendix 7B -Geotechnical Infrastructure"

d) Annexure – 1 (List of tests proposed for Geotechnical Investigation)

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6.0 ANNEXURE 1

ANNEXURE 1

LIST OF TESTS PROPOSED FOR GEOTECHNICAL INVESTIGATION

S. No.	Name of Test	Clause No. (vide Ma'aden Specification #MD-101-SMEM-EG-CV-SPC-0009)	Purpose of test	No. of tests	Design Parameters obtained from test output
1	Boreholes	7.5	This test is used to determine the sub-surface characteristic.	29	Soil profile
2	Trial Pits	7.5	This test is done for visual examination of soil profile and characteristic. Sample is collected and geotechnical parameters can be obtained from lab tests on sample.	50	Soil profile, visual examination
3	Standard Penetration Test (SPT)	8.1	This test is used to determine the sub-surface characteristic.	-	SPT 'N' value
4	Electrical Resistivity Test	8.8.1	Soil resistivity testing is the processes of measuring a volume of soil to determine the conductivity of the soil.	16	Soil resistivity (R)
5	Piezometer test	9.1	This test is done to determine the ground water level	3	Groundwater level
6	Static Plate Load Test	-	This test is done to determine the bearing capacity of soil/rock	7	Bearing capacity
7	Environmental boreholes	-	This test is done to determine the chemical content of soil/rock and water	3	Chemical content
8	Downhole seismic test	-	This test is done to determine the seismic wave velocity in soil/rock	3	Coeff. of elastic uniform and

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S. No.	Name of Test	Clause No. (vide Ma'aden Specification #MD-101-SMEM-EG-CV-SPC-0009)	Purpose of test	No. of tests	Design Parameters obtained from test output
					non-uniform compression (C_u & C_ϕ), coeff. of elastic uniform and non-uniform shear (C_ζ & C_ψ), elastic modulus (E), shear modulus (G)
9	Gradation analysis (sieve analysis and hydrometer analysis)	10.2	This test is done to determine particle size distribution curve	29	Particle size distribution
10	Liquid, plastic and shrinkage limits	10.2	This test is done to determine consistency index	29	Liquid limit, plastic limit and shrinkage limit
11	Specific gravity	10.2	This test is done to determine specific gravity	29	Specific gravity
12	Chemical analysis	10.2	This test is done to determine chemical content	29	Chemical content
13	Proctor compaction	10.2	This test is done to determine optimum moisture content	29	Optimum moisture content
14	California bearing ratio	10.2	This test is done to determine CBR value	19	CBR value
15	Bulk density	10.2	This test is done to determine bulk	29	Bulk density

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S. No.	Name of Test	Clause No. (vide Ma'aden Specification #MD-101-SMEM-EG-CV-SPC-0009)	Purpose of test	No. of tests	Design Parameters obtained from test output
	and moisture content		density and moisture content		and moisture content
16	Relative density	10.2	This test is done to determine relative density	29	Relative density
17	Direct shear test	10.2	This test is done to determine shear strength parameters	29	Cohesion and friction angle
18	Permeability	10.2	This test is done to determine coefficient of permeability	12	Coeff. of permeability
19	Water absorption, porosity and density test	10.2	This test is done to determine water absorption value, porosity and density	58	Water absorption, porosity and density
20	Specific gravity	10.2	This test is done to determine specific gravity	87	Specific gravity
21	Unconfined compression	10.2	This test is done to determine shear strength parameters	87	Rock strength
22	Point load strength index	10.2	This test is done to determine shear strength parameters	58	Rock strength
23	Thermal Resistivity Test (TRT)	-	This test is done to determine the capacity of ground to conduct or dissipate heat	1	Thermal resistivity

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7.0 ATTACHMENT 1

Comment Resolution Log Sheet / Compliance Report

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BECHTEL

Mansourah Massarah Gold Project
Comment Resolution Log Sheet / Compliance Report

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PROJECT: MANSOURAH MASSARAH GOLD PROJECT — **PROJECT NO.: 216**

STAGE: 4 - PROJECT EXECUTION. — **REVIEW LOG DOC. NO. & REV.:**

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Mansourah Massarah Gold Project
Comment Resolution Log Sheet / Compliance Report

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PROJECT: MANSOURAH MASSARAH GOLD PROJECT **PROJECT NO.: 216**

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